

Some Unpleasant Monetarist Arithmetic

Sargent, Wallace 1981; Werning 2021

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Main claims

Friedman's 1968 warning:

- Monetary policy cannot permanently control real output, unemployment, or real returns.
- But it should be able to control short and long term inflation, at least

This paper challenged even that: can it really?

- If fiscal policy is stabilizing, yes
- However, under the "right" fiscal setup, not even full on Friedman monetarism saves monetary policy

The Unpleasant Result

Under fiscal dominance, monetary arithmetic can spectacularly defeat monetary intent.

Microfoundations at a glance

The main text is intentionally reduced-form. Microfoundations are relegated to the appendix

- The economy is an OLG endowment one
- Agents live for two periods
 - There are poor and rich agents
 - Endowments arrive only when young at first; later, on both periods
- Poor agents can save only through currency, generating money demand
- Rich ones can save through bonds or capital.
- The government finances itself via interest-bearing real bonds and base money.

Segmentation $\Rightarrow$ money demand $\Rightarrow$ seigniorage revenue.

Results generated by the model

From now on, we will focus on the following properties:

- ① **Real income grows at rate n .**
 - ② **Government securities pay a constant real return R , with $R > n$.**
 - ③ **Base-money demand obeys the quantity theory:** constant income velocity.
 - ④ **There is an upper bound on real government debt** the private sector is willing to absorb.
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- **This is a strongly monetarist environment.** The price level is tightly linked to the monetary base.
 - **Indexation is not central here.** The model can be read in nominal terms as long as agents are not systematically wrong about inflation.

Policies in the model

- **Fiscal policy is exogenous.** The deficit sequence $\{D_t\}$ is independent
- Monetary policy is a money-growth rule.

$$H_t = (1 + \theta)H_{t-1}, \quad t < T.$$

A lower θ means tighter money.

- Per capita debt evolves according to:

$$\frac{B_t}{N_t} = \underbrace{\frac{(1 + R_{t-1})}{(1 + n)} \cdot \frac{B_{t-1}}{N_{t-1}}}_{\text{interest bill}} + \underbrace{\frac{D_t}{N_t}}_{\text{new deficit}} - \underbrace{\frac{H_t - H_{t-1}}{N_t p_t}}_{\text{seigniorage}}$$

- **Debt eventually binds** when it reaches its upper bound. We call this period T
- After T , monetary policy must switch to whatever money growth stabilizes debt

Result 1: lower θ today, higher inflation tomorrow

Setup

Choosing monetary policy equals choosing inflation given quantity-theory

$$p_t = \frac{1}{h} \frac{H_t}{N_t} \quad \Rightarrow \quad \frac{p_t}{p_{t-1}} = \frac{1 + \theta}{1 + n}.$$

- Before T , monetary policy is free
- When the debt ceiling binds ($b = b_\theta(T)$) it must generate enough inflation to stabilize it
- Inflation after T depends on $b_\theta(T)$:

$$1 - \frac{p_{t-1}}{(1+n)p_t} = \frac{1}{h} \left[\frac{D_t}{N_t} + \frac{(R-n)b_\theta(T)}{1+n} \right], \quad t > T$$

$\Rightarrow$ Higher debt requires higher inflation

Result 1: Lower θ today, higher inflation tomorrow

Result

It can be shown that lower θ before T raises $b_\theta(T)$.

- The math is... unpleasant
- But the result itself is very intuitive!

$\theta \downarrow \Rightarrow$ Less seigniorage and less inflation $\Rightarrow$ More debt emissions $\Rightarrow b_\theta(T) \uparrow$

- At T debt must be stabilized:

No more emissions $\Rightarrow$ More seigniorage is needed; $\theta \uparrow \Rightarrow \frac{p_t}{p_{t-1}} \uparrow$

Lower θ today achieves lower inflation today, **but requires higher inflation after T**

Result 2: lower θ today, higher inflation *today*

Setup

The previous result uses a simple money demand schedule

- What if money demand is forward looking?
- Then expected accommodation feeds back into today

A note on microfoundations:

- No old-age endowments generates the simple money demand
- Old-age endowments to poor agents changes that. Now we get:

$$\frac{H_t}{N_t p_t} = \frac{\gamma_1}{2} - \frac{\gamma_2}{2} \frac{p_{t+1}}{p_t}$$

- Expected loose money in old age means higher p_{t+1}
- N_t is exogenous and H_t is fixed by policy
- So left hand adjustments must happen via higher p_t

Result 2: lower theta today, higher inflation *today*

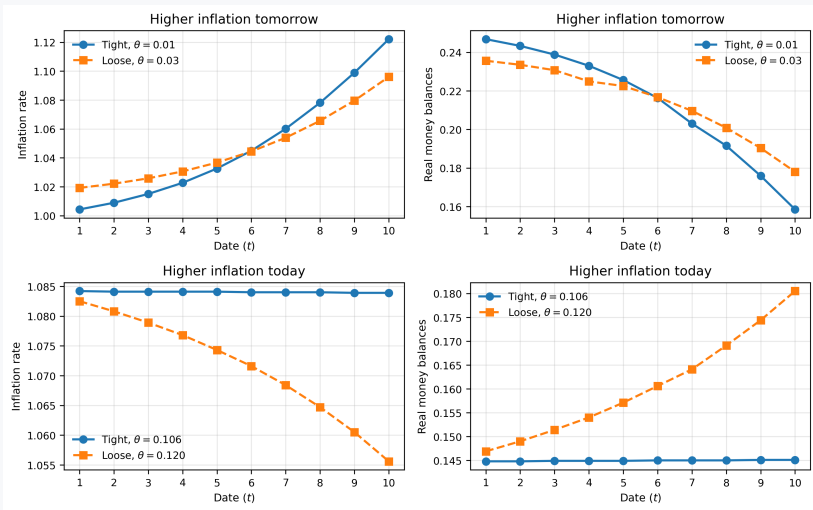
Inflation dynamics

Solving the new money demand forward, we get the new price dynamics:

$$p_t = \frac{2}{\gamma_1} \sum_{j=0}^{\infty} \left[\left(\frac{\gamma_2}{\gamma_1} \right)^j \frac{H_{t+j}}{N_{t+j}} \right].$$

- **Current inflation** now depends on the **entire expected money path**.
- The authors then proceed to simulate this economy

Intermediate and spectacular examples of the effects of monetary policy



Conditions for the truly unpleasant result

The spectacular result is not generic. It requires a restrictive region of the parameter space.

- **The economy must be fiscally fragile.**
 - Debt service must raise future money creation enough that the expectation channel dominates the money channel for inflation
- **Poor agents old-age endowments must be high .**
 - γ_1 and γ_2 must be close
 - And γ_2 must be high

Economically, this means the result is attained when:

- The poor population is rich in real terms
- Is still disallowed from investing
- Are as rich in old age as in their youth

Recap

Recapitulating, when accounting for fiscal policy:

- A simple and orthodox monetarist model weakens monetary control of inflation.
- A slightly richer and "well calibrated" one completely undoes it

The mechanism requires three key ingredients:

- Some agents must hold money rather than bonds;
- The government must collect seigniorage.
- Fiscal policy must be independent and unconstrained
 - The adjustment burden falls on monetary policy

No one cares about money growth anymore

Monetary policy is done with interest rates, not money growth

- Werning's question: can the Sargent–Wallace arithmetic be restated using interest rates?
- Yes!
- The result becomes a public-finance statement
- It clarifies how nominal interest rates are a tax on money balances

The model redux

Werning reduces Sargent and Wallace's model to three equations:

Budget constraint: $B_{t,t+1} + M_{t,t+1} = P_t d_t + (1 + i_t)B_{t-1,t} + M_{t-1,t},$

Money demand: $\frac{M_{t,t+1}}{P_t} = L(i_{t+1}),$

Fisher equation: $1 + i_t = (1 + r_t)(1 + \pi_t).$

- Real rate is exogenous; fiscal deficits are real, and money demand is non increasing in the nominal rate.
- Debt is nominal. The real value of inherited debt is pinned down by the price level.
- Sargent–Wallace work with real debt!

Working in Present Value

Let $q_t \equiv \frac{1}{(1+r_1)\dots(1+r_t)}$, and we get

$$\underbrace{\sum_{t=0}^{\infty} q_t \frac{M_{t,t+1} - M_{t-1,t}}{P_t}}_{\text{PV of seigniorage}} - \underbrace{\frac{(1+i_0)B_{-1,0}}{P_0}}_{\text{Initial real debt}} = \underbrace{\sum_{t=0}^{\infty} q_t d_t}_{D_t; \text{ PV of future deficits}}$$

Equivalently,

$$\sum_{t=0}^{\infty} q_t \frac{i_t}{1+i_t} L(i_{t+1}) - \omega L(i_1) = D_t; \quad \omega > 0$$

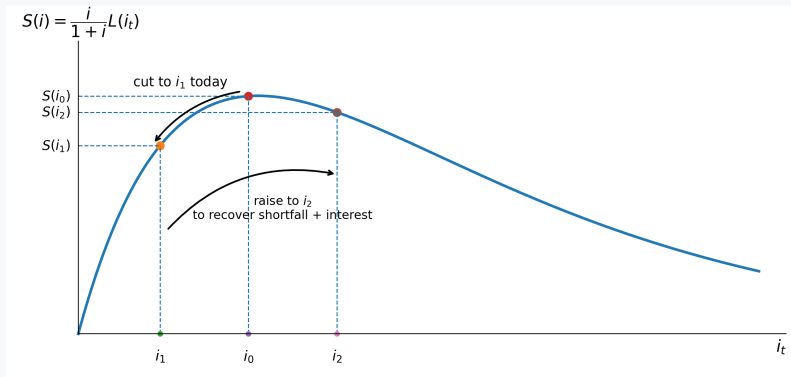
$\Rightarrow \frac{i}{1+i} L(i)$ reveals a Laffer curve for the money-tax revenue!

Equivalency

Werning argues there is a 1:1 mapping between this setup and Sargent–Wallace's:

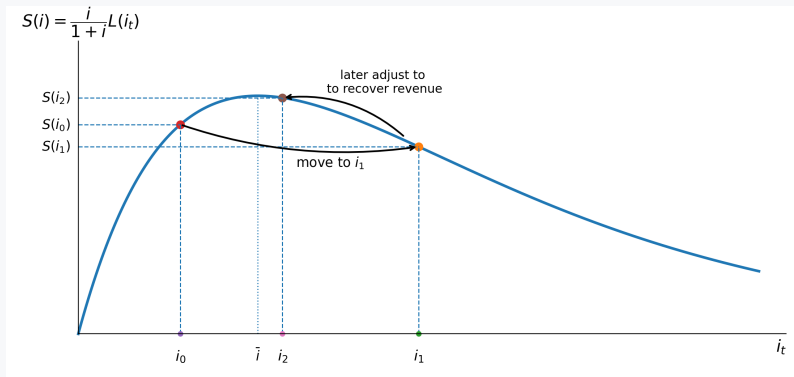
- A nominal-rate sequence implies a unique path for prices and money under fixe M_0
- The reverse mapping also holds under convergence
- **Therefore the same policy experiments can be expressed in either language.**

Unpleasant result 1: higher inflation tomorrow



- On the good side of the Laffer curve, lower rates lower money-tax revenue
- Present-value revenue must remain unchanged; Rates eventually rise.
- The Fisher equation maps higher nominal rates into higher inflation.

Unpleasant Result 2: Higher inflation always



- The stronger result uses the bad side of the Laffer curve.
- Fisher does the rest. With real rates fixed, higher nominal rates mean higher inflation.

Taking stock

- Werning's exercise reinterprets the classical result in terms of interest rates
- It clarifies how it is actually a public finance result
- And makes more explicit the channel through which policy backfires
- Lastly, it showcases how the relation between monetary growth, interest rates and inflation can be complicated

A pithy takeaway, given this is a fiscal topics class:

Monetary policy cannot be separated from fiscal financing.

Bonus: Unpleasant New Keynesian Arithmetic?

Can we port these results to the canonical NK model?

- The answer depends on what we mean by “these result.”
- The original model requires some agents hold debt and the government collects seigniorage.
- **The canonical NK model is representative agent and cashless.**
- **So the mechanism cannot be identical.**
- We must rely only on debt revaluation $\rightarrow$ debt must be nominal.

Why would we want to?

- The original model is pure endowments, no production
- No one believes in a stable money demand schedule
- Thinking about monetary policy only through a Fisher equation gives me headaches

A limited equivalence

A NK setting can never deliver direct monetary financing. There is no money!
However, with nominal debt:

- Inflation can reduce the real value of inherited liabilities
- This is the FTPL channel re-stated
- There is no Laffer curve for money-tax revenue because there is no money
- **But we can ask a parallel question.** Can an unbacked fiscal deficit make tighter monetary policy inflationary tomorrow? And today?

It can. In fact the second result is easier, and standard FTPL

- Achieving the first result is the hard part
- But as with Werning, it is also quite elucidative

One last model

Consider the standard NK model:

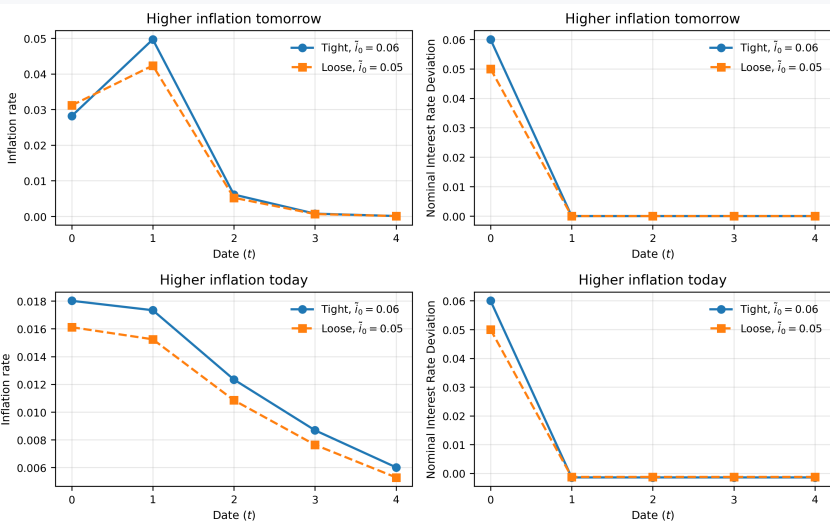
$$\text{IS : } \tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \quad (1)$$

$$\text{NKPC : } \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \quad (2)$$

$$\text{Gov. Budget : } b_t = \frac{1}{\beta} (b_{t-1} + b^{ss} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) - s_t) \quad (3)$$

- **Fiscal experiment:** $s_0 = -d_0 < 0$, and $s_t = 0$ for all $t > 0$.
- FTPL logic tells us i_0 has to adjust lower to deliver equilibrium. What if it doesn't?

Unpleasant New Keynesian simulations!



A big caveat: is this credible?

The result is reproducible, but knife-edge.

- The immediate-inflation result is just a "one-period lagged" FTPL
 - The CB tries to resist, but equilibrium selection forces its hand
- **The delayed-inflation result is much trickier**
- The recession induced by higher rates must overpower the fiscal revaluation channel
- That requires non plausible calibration: $\kappa > 1$ and $\sigma < 1$
- That is, marginal costs move strongly with output gap. The NKPC is steep.
- And demand is very rate-sensitive. Agents are not too risk averse and have high IES
- Sargent–Wallace require rich-but-constrained agents; Werning relies entirely on Fisher

Thank you!

Questions and comments are very welcome